

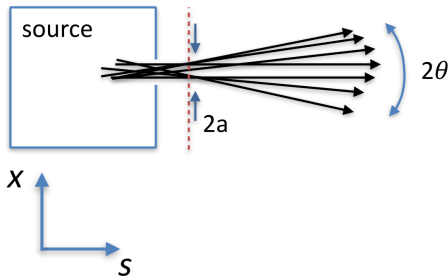
Courant-Snyder Parametrization, Beam Mismatch, Normalized Emittance

Particle beams

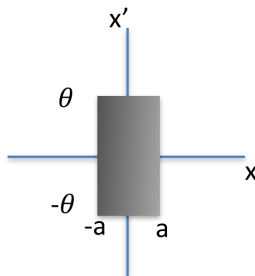
- ▶ Not just one particle, but an ensemble (bunch, group) of particles that stay together for the duration of observation
- ▶ Finite spread in particle properties
 - ▶ energy/momentum spread
 - ▶ position/direction spread
- ▶ Characterization in terms of “phase space”
- ▶ Adiabatic invariance of phase space variables
 - ▶ position/momentum; energy/time

$$x' = \frac{dx}{ds}$$

Physical space:

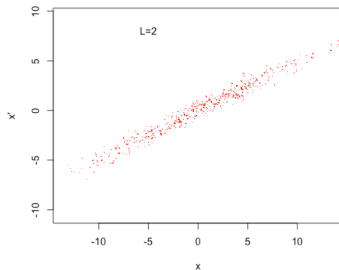
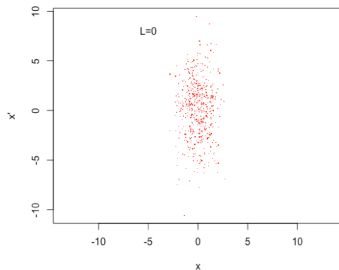


Phase space:



Evolution of the Phase Space

- ▶ Suppose transverse positions, x , and transverse angles, x' , are distributed in a normal (Gaussian) fashion coming from the “source”
- ▶ If the distribution is allowed to “drift” a distance L , then $x = x_0 + Lx'_0$ for each particle; the particles with larger $x' = x'_0$ will quickly drift to larger x values, and the distribution will “shear”
- ▶ Shape and orientation of distribution in “phase space” will change, but effective “area” of distribution will remain constant!

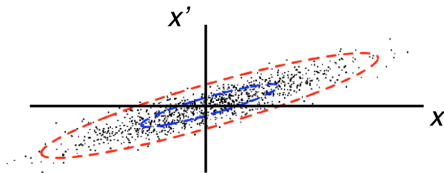


Emittance in terms of moments of the distribution

- ▶ Considering the general equation of an ellipse, the area enclosed by the ellipse is related to the coefficients by

$$ax^2 + bxx' + cx'^2 = 1$$

$$\mathcal{A} = \frac{2\pi}{\sqrt{4ac - b^2}}$$



- ▶ Details are here: <http://nicadd.niu.edu/~syphers/uspas/2018w/some-notes-on-ellipses.html>
- ▶ Can define scaled quantities from our distribution:

$$\alpha = -\frac{\langle xx' \rangle}{\varepsilon/\pi}, \quad \beta = \frac{\langle x^2 \rangle}{\varepsilon/\pi}, \quad \gamma = \frac{\langle x'^2 \rangle}{\varepsilon/\pi}, \quad \text{where } \boxed{\varepsilon = \pi \sqrt{\langle x^2 \rangle \langle x'^2 \rangle - \langle xx' \rangle^2}}$$

is the **rms emittance**, and α, β, γ are Courant-Snyder or Twiss parameters

- ▶ The equation of the **blue** ellipse is $\boxed{\gamma x^2 + 2\alpha xx' + \beta x'^2 = \varepsilon/\pi}$
- ▶ **Red** ellipse contains $\sim 95\%$ particles and has an area of $\sim 6\varepsilon$

Beam transport and focusing

- ▶ Consider a section of finite length containing pure uniform magnetic field to bend a charged particle's trajectory through a portion of a circular arc, thus steering it in a new direction.
- ▶ An arrangement of such magnets can be used to guide an “ideal” particle from one point to another
- ▶ However, most particles are NOT ideal! Hence, as particles drift away from the ideal trajectory, we need to guide them (using quadrupoles or solenoids) back toward the ideal.
- ▶ We use discrete electromagnets of finite length and assume a linear relationship between a particle's exit trajectory to its entrance trajectory, depending upon the strength of the magnetic field.
 - ▶ (we will address nonlinear effects later to some extent)

Linear optics

- ▶ Let x be the transverse displacement of a particle from the ideal beam trajectory. Let the angle it makes to the ideal trajectory be $x' = dx/ds$, where s is the distance (arclength) along the ideal trajectory. [Linear] transport through a magnetic element is then described by a matrix M , such that

$$\mathbf{X} = M\mathbf{X}_0, \quad \mathbf{X} = \begin{pmatrix} x \\ x' \end{pmatrix}$$

- ▶ An arbitrary trajectory, relative to the design trajectory, can be computed via matrix multiplication for elements all along the beam line:

$$\begin{pmatrix} x_N \\ x'_N \end{pmatrix} = M_N M_{N-1} \dots M_2 M_1 \begin{pmatrix} x_0 \\ x'_0 \end{pmatrix}$$

- ▶ Note: sometimes x_0 is denoted as x_i (initial) and x_N as x_f (final)

Matrix formalism

- ▶ Write solution to each “piece” of the beam transport system in matrix form
 - ▶ for each piece, assume $K = \text{const}$ from $s = 0$ to $s = L$

- ▶ Free space/drift:

$$\begin{pmatrix} x \\ x' \end{pmatrix} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x'_0 \end{pmatrix}$$

- ▶ Focusing quadrupole, $K > 0$:

$$\begin{pmatrix} x \\ x' \end{pmatrix} = \begin{pmatrix} \cos(\sqrt{K}L) & \frac{1}{\sqrt{K}} \sin(\sqrt{K}L) \\ -\sqrt{K} \sin(\sqrt{K}L) & \cos(\sqrt{K}L) \end{pmatrix} \begin{pmatrix} x_0 \\ x'_0 \end{pmatrix}$$

- ▶ Defocusing quadrupole, $K < 0$:

$$\begin{pmatrix} x \\ x' \end{pmatrix} = \begin{pmatrix} \cosh(\sqrt{|K|}L) & \frac{1}{\sqrt{|K|}} \sinh(\sqrt{|K|}L) \\ \sqrt{|K|} \sinh(\sqrt{|K|}L) & \cosh(\sqrt{|K|}L) \end{pmatrix} \begin{pmatrix} x_0 \\ x'_0 \end{pmatrix}$$

- ▶ Note: $\boxed{\det M = 1}$

Thin lens quadrupole

- ▶ If a quadrupole magnet is short enough, particle's offset through the quad does not change by much, but the slope of the trajectory does
 - ▶ acts like a “thin lens” in geometrical optics
- ▶ Take limit as $L \rightarrow 0$, while KL remains finite:

$$\begin{pmatrix} \cos(\sqrt{K}L) & \frac{1}{\sqrt{K}} \sin(\sqrt{K}L) \\ -\sqrt{K} \sin(\sqrt{K}L) & \cos(\sqrt{K}L) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -KL & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix}$$

- ▶ $KL = \frac{1}{f} = \frac{B'L}{B\rho}$
- ▶ You can use the same approach for a defocusing quad
- ▶ Approximation is valid if $f \gg L$

Transport of beam moments

- ▶ Transport of particle state vector downstream from initial position:

$$\mathbf{X} = M \mathbf{X}_0$$

- ▶ Corresponding covariance matrix of the resulting vector:

$$\mathbf{X} \mathbf{X}^T = \begin{pmatrix} x^2 & x x' \\ x' x & x'^2 \end{pmatrix} = M \mathbf{X}_0 (M \mathbf{X}_0)^T = M \mathbf{X}_0 \mathbf{X}_0^T M^T$$

- ▶ Then, by averaging over all the particles in the distribution,

$$\Sigma = \begin{pmatrix} \langle x^2 \rangle & \langle x x' \rangle^2 \\ \langle x x' \rangle^2 & \langle x'^2 \rangle \end{pmatrix}, \text{ we get } \boxed{\Sigma = M \Sigma_0 M^T}$$

Transport of beam moments II

- ▶ Since $\Sigma = \begin{pmatrix} \langle x^2 \rangle & \langle xx' \rangle^2 \\ \langle xx' \rangle^2 & \langle x'^2 \rangle \end{pmatrix} \begin{pmatrix} \varepsilon\beta & -\varepsilon\alpha \\ -\varepsilon\alpha & \varepsilon\gamma \end{pmatrix} = \varepsilon K$,
- ▶ where $K = \begin{pmatrix} \beta & -\alpha \\ -\alpha & \gamma \end{pmatrix}$,
- ▶ then $\boxed{K = MK_0M^T}$
- ▶ If the transfer matrix M is known, we can “transport” beam distribution parameters from one point to any point downstream, which determines beam distribution along the way:

$$\beta = \frac{\langle x^2 \rangle}{\varepsilon/\pi} \Rightarrow x_{\text{rms}} = \sqrt{\varepsilon\beta(s)/\pi}$$

Conservation of emittance

- ▶ From $\Sigma = M\Sigma_0M^T$, where $\Sigma = \varepsilon K$, $K = \begin{pmatrix} \beta & -\alpha \\ -\alpha & \gamma \end{pmatrix}$,
- ▶ it follows that $\det \Sigma = \varepsilon^2 \det K = \varepsilon^2(\beta\gamma - \alpha^2) = \varepsilon^2$
- ▶ The emittance is conserved upon transport through the system

Numbers and units

$$\varepsilon = \pi \sqrt{\langle x^2 \rangle \langle x'^2 \rangle - \langle xx' \rangle^2}$$

- ▶ If $\langle x^2 \rangle \sim \text{mm}^2$, and $\langle x'^2 \rangle \sim \text{mrad}^2$, then the emittance has units of $\text{mm} \cdot \text{mrad}$ (also = μm)
- ▶ Courant-Snyder parameters

$$\beta = \frac{\pi \langle x^2 \rangle}{\varepsilon} \quad \text{mm}^2/(\text{mm} \cdot \text{mrad}) \sim \text{mm}/\text{mrad} = \text{m}$$

$$\alpha = -\frac{\pi \langle xx' \rangle}{\varepsilon} \quad \text{mm} \cdot \text{mrad}/(\text{mm} \cdot \text{mrad}) = \text{dimensionless}$$

$$\gamma = \frac{\pi \langle x'^2 \rangle}{\varepsilon} \quad \text{mrad}^2/(\text{mm} \cdot \text{mrad}) = 1/\text{m}$$

- ▶ The “ π ” comes from our definition of emittance as an area in phase space; emittance is often expressed in units of “ $\pi \text{ mm} \cdot \text{mrad}$ ”

Summary

- ▶ Given an initial particle distribution in phase space at the input to a beam transport system, can describe that distribution (sometimes not all that well, but we try...) using Courant-Snyder parameters:

$$\varepsilon = \pi \sqrt{\langle x^2 \rangle \langle x'^2 \rangle - \langle xx' \rangle^2}$$

$$\beta = \frac{\pi \langle x^2 \rangle}{\varepsilon}$$

$$\alpha = -\frac{\pi \langle xx' \rangle}{\varepsilon}$$

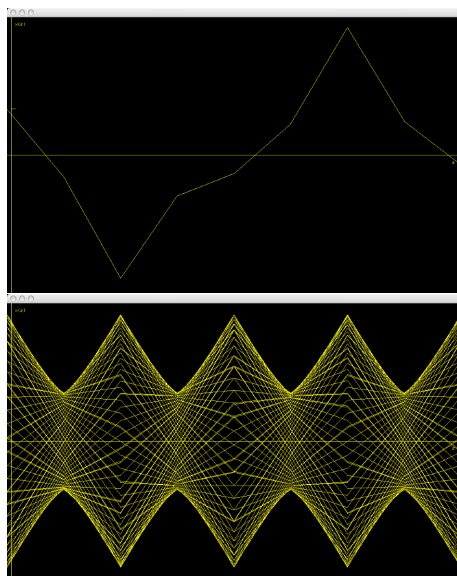
$$\gamma = \frac{\pi \langle x'^2 \rangle}{\varepsilon}$$

- ▶ The C-S parameters can then be computed downstream (propagated through our system), using

$$\Sigma = M \Sigma_0 M^T$$

Amplitude function

- ▶ Can trace single particle trajectories through a periodic system
- ▶ It can represent either
 - ▶ multiple passages around a circular accelerator, or
 - ▶ multiple particles through a beam line
- ▶ Can we describe the maximum amplitude of particle excursions in analytical form?
- ▶ Yes! Stay tuned...



Hill's Equation — Analytical Solution

- We saw that the equation of transverse motion is Hill's Equation:

$$x'' + K(s)x = 0$$

- Note: “similar” to simple harmonic oscillator equation, but “spring constant” is not *constant* -- depends upon longitudinal position, s .
- So, assume solution is sinusoidal, with a phase which advances as a function of location s ; also assume amplitude is modulated by a function which also depends upon s :

$$x(s) = A\sqrt{\beta(s)} \sin[\psi(s) + \delta]$$

- Then, plug into Hill's Equation ...



Analytical Solution (cont'd)

$$x(s) = A\sqrt{\beta(s)} \sin[\psi(s) + \delta]$$

$$x' = \frac{1}{2}A\beta^{-\frac{1}{2}}\beta' \sin[\psi(s) + \delta] + A\sqrt{\beta} \cos[\psi(s) + \delta]\psi'$$

$$x'' = \dots$$

Plugging into Hill's Equation, and collecting terms...

$$\begin{aligned} x'' + K(s)x &= A\sqrt{\beta} \left[\psi'' + \frac{\beta'}{\beta}\psi' \right] \cos[\psi(s) + \delta] \\ &\quad + A\sqrt{\beta} \left[-\frac{1}{4} \frac{(\beta')^2}{\beta^2} + \frac{1}{2} \frac{\beta''}{\beta} - (\psi')^2 + K \right] \sin[\psi(s) + \delta] = 0 \end{aligned}$$

A and δ are constants of integration, defined by the initial conditions (x_0, x'_0) of the particle. For arbitrary A , δ , must have contents of each $[] = 0$ simultaneously for $sum = 0$.



Analytical Solution (cont'd)

$$x(s) = A\sqrt{\beta(s)} \sin[\psi(s) + \delta]$$

thus, we need

$$\psi'' + \frac{\beta'}{\beta} \psi' = 0$$

$$\beta \psi'' + \beta' \psi' = 0$$

$$(\beta \psi')' = 0$$

$$\beta \psi' = \text{const}$$

$$\psi' = 1/\beta$$

and

$$-\frac{1}{4} \frac{(\beta')^2}{\beta^2} + \frac{1}{2} \frac{\beta''}{\beta} - (\psi')^2 + K = 0$$

$$2\beta\beta'' - (\beta')^2 - 4\beta^2(\psi')^2 + 4K\beta^2 = 0$$

$$2\beta\beta'' - (\beta')^2 + 4K\beta^2 = 4$$

Note: the phase advance is an observable quantity. So, while could choose different value of *const*, then β would just scale accordingly; thus, valid to choose *const* = 1.

The function $\beta(s)$ is the local wavelength ($\lambda/2\pi$) of the oscillatory motion.

Differential equation that the amplitude function must obey

Comments on Hill's equation solution

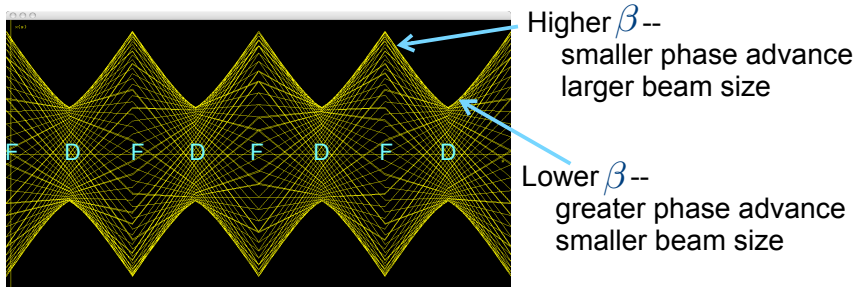
$$x(s) = A\sqrt{\beta(s)} \sin(\psi(s) + \delta)$$

- ▶ We chose the amplitude function to be a positive definite function in its definition, since we want to describe real solutions.
- ▶ The square root of the amplitude function determines the shape of the envelope of a particle's motion. But the amplitude function is also a local wavelength of the motion.
- ▶ This seems strange at first, but...
 - ▶ Consider a particle oscillating within our focusing lens system; if the lenses are suddenly spaced further apart, the particle's motion amplitude will grow larger between lenses, and additionally it will travel further before a complete oscillation takes place. If the lenses are spaced closer together, the oscillation will not be allowed to grow as large, and more oscillations will occur per unit distance traveled.
 - ▶ Thus, the spacing and/or strengths (i.e., $K(s)$) determine both the rate of change of the oscillation phase as well as the maximum oscillation amplitude. These attributes must be tied together.

The Amplitude Function, β

$$x(s) = A\sqrt{\beta(s)} \sin[\psi(s) + \delta]$$

- Since the amplitude function is a wavelength, it might have numerical values of many meters, say. However, typical particle transverse motion is on the scale of mm. So, this means that the constant A must have units of $\text{m}^{1/2}$, and it must be numerically small. More on this subject coming up...



Equation of motion of amplitude function

$$2\beta\beta'' - (\beta')^2 + 4K\beta^2 = 4$$

Differentiate once:

$$2\beta'\beta'' + 2\beta\beta''' - 2\beta'\beta'' + 4K'\beta^2 + 8K\beta\beta' = 0$$

$$\beta''' + 4K\beta' + 2K'\beta = 0$$

Typically, $K'(s) = 0$ (think drift or hard-edge quadrupole), so

$$(\beta'' + 4K\beta)' = 0, \quad \beta'' + 4K\beta = \text{const}$$

is the general equation of motion for the amplitude function, β (in regions where K is either zero or constant).

If $K = 0$, $\Rightarrow$

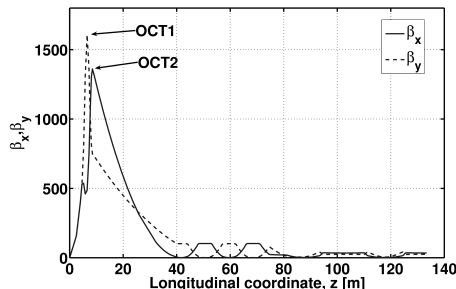
$$\beta'' = \text{const}, \Rightarrow \beta(s) = \beta_0 + \beta'_0 s + \frac{1}{2}\beta''_0 s^2,$$

also

$$\beta > 0, \Rightarrow 2\beta\beta'' - (\beta')^2 = 4 \text{ yields } \beta'' = \frac{4 + (\beta')^2}{2\beta} > 0$$

The trajectory is a parabola, concave up!

Amplitude function, β



- ▶ Since the amplitude function is a wavelength, it might have numerical values of many meters.
 - ▶ Shown on the left is a β -function plot for the interaction region of some collider design.
 - ▶ Since the particles need to be squeezed very tight, the coordinates are very small, but the angles are large.
 - ▶ β shows its typical parabolic shape (more in the next slide).
- ▶ However, typical particle transverse motion is on the scale of mm, $\Rightarrow$ constant A must have units of $m^{1/2}$, and it must be numerically small.

Courant-Snyder Parameters, & Connection to Matrix Approach

- Suppose, for the moment, that we know the value of the amplitude function and its slope at two points along our particle transport system.
- Previously have seen how to write the motion of a single particle in one degree of freedom between two points in terms of a matrix. We can now recast the elements of this matrix in terms of the local values of the amplitude function.
- Define two new variables, $\alpha \equiv -\frac{1}{2}\beta', \quad \gamma \equiv \frac{1 + \alpha^2}{\beta}$
- Collectively, β, α, γ are called the *Courant-Snyder Parameters* (sometimes called “Twiss parameters” or “lattice parameters”)

$$2\beta\beta'' - (\beta')^2 + 4K\beta^2 = 4 \quad \text{becomes} \quad K\beta = \gamma + \alpha'$$

Solutions using Courant-Snyder Parameters

- Our previous results become

▸ drift space:

$$\beta(s) = \beta_0 + \beta'_0 s + \frac{1}{2} \beta''_0 s^2$$

$$\rightarrow \beta(s) = \beta_0 - 2\alpha_0 s + \gamma_0 s^2$$

- gradient field:

$$\beta(s) = \beta_0 + \frac{\beta'_0}{2\sqrt{K}} \sin(2\sqrt{K}s) + \frac{\beta''_0}{4K} [1 - \cos(2\sqrt{K}s)]$$

$$\rightarrow \beta(s) = \frac{\beta_0}{2} [1 + \cos(2\sqrt{K}s)] - \frac{\alpha_0}{\sqrt{K}} \sin(2\sqrt{K}s) + \frac{\gamma_0}{2K} [1 - \cos(2\sqrt{K}s)]$$

The Transport Matrix

- We can always write:

$$x(s) = a\sqrt{\beta} \sin \Delta\psi + b\sqrt{\beta} \cos \Delta\psi$$

- Solve for a and b in terms of initial conditions and write in matrix form

- ▶ we get:

$$\begin{pmatrix} x \\ x' \end{pmatrix} = \begin{pmatrix} \left(\frac{\beta}{\beta_0}\right)^{1/2} (\cos \Delta\psi + \alpha_0 \sin \Delta\psi) & \sqrt{\beta_0\beta} \sin \Delta\psi \\ -\frac{1+\alpha_0\alpha}{\sqrt{\beta_0\beta}} \sin \Delta\psi - \frac{\alpha-\alpha_0}{\sqrt{\beta_0\beta}} \cos \Delta\psi & \left(\frac{\beta_0}{\beta}\right)^{1/2} (\cos \Delta\psi - \alpha \sin \Delta\psi) \end{pmatrix} \begin{pmatrix} x_0 \\ x'_0 \end{pmatrix}$$

So, can write any of our transport matrices in terms of values of C-S parameters at the two end points, and the phase advance between them.

$\Delta\psi$ is the phase advance from point s_0 to point s in the beam line

Tracking β , α , γ ...

- Saw earlier that if given values of the Courant-Snyder parameters at one location in the beam line, and if know the matrix of the linear motion between that location and another location downstream, then can compute the values at the second location *via*:

$$K = M K_0 M^T$$

where $K \equiv \begin{pmatrix} \beta & -\alpha \\ -\alpha & \gamma \end{pmatrix}$

- Have not explicitly proven that the ellipse coefficients found earlier are the SAME as the parameters above, but they are — and, we will.

Evolution of the Phase Advance

- Also, if know parameters at one point, and the matrix from there to another point, then

$$M_{1 \rightarrow 2} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \frac{b}{a\beta_1 - b\alpha_1} = \tan \Delta\psi_{1 \rightarrow 2}$$

- So, from knowledge of matrices, can “transport” phase *and* the Courant-Snyder parameters along a beam line from one point to another

Simple Examples

- Propagation through a Drift:

$$M = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \Delta\psi = \tan^{-1} \left(\frac{L}{\beta_1 - L\alpha_1} \right)$$

$$\beta = \beta_0 - 2\alpha_0 L + \gamma_0 L^2$$

$$\alpha = \alpha_0 - \gamma_0 L$$

$$\gamma = \gamma_0$$

- Propagation through a Thin Lens:

$$M = \begin{pmatrix} 1 & 0 \\ -1/F & 1 \end{pmatrix}$$

$$\Rightarrow \Delta\psi = 0$$

$$\beta = \beta_0$$

$$\alpha = \alpha_0 + \beta_0/F$$

$$\gamma = \gamma_0 + 2\alpha_0/F + \beta_0/F^2$$

$$K = M K_0 M^T$$

- Given α, β at one point, can calculate α, β at all downstream points

Betatron Oscillation Amplitude

- Transverse oscillations in a synchrotron (or beam line) are called Betatron Oscillations (first observed/analyzed in a “betatron” accelerator)
- Write x, x' in terms of initial conditions x_0, x'_0 :

$$x(s) = a\sqrt{\beta} \cos \Delta\psi + b\sqrt{\beta} \sin \Delta\psi$$

$$x' = \frac{1}{\sqrt{\beta}} ([b - a\alpha] \cos \Delta\psi - [a + b\alpha] \sin \Delta\psi)$$

↓

$$a = \frac{x_0}{\sqrt{\beta_0}}, \quad b = \frac{\alpha_0 x_0 + \beta_0 x'_0}{\sqrt{\beta_0}}$$

$$\Rightarrow x(s) = \sqrt{\frac{\beta(s)}{\beta_0}} [x_0 \cos \Delta\psi + (\alpha_0 x_0 + \beta_0 x'_0) \sin \Delta\psi]$$

$$\text{amplitude: } A = \sqrt{\frac{x_0^2 + (\alpha_0 x_0 + \beta_0 x'_0)^2}{\beta_0}}$$

Free Betatron Oscillation

- Suppose a particle traveling along the design path is given a sudden (impulse) deflection through angle $\Delta x' = x'_0 = \Delta\theta$
- Then, downstream, we have

$$x(s) = \Delta\theta \sqrt{\beta_0 \beta(s)} \sin[\psi(s) - \psi_0]$$



Example:

Suppose $\Delta\theta = 0.4$ mrad, $\beta_0 = 4.0$ m, $\beta(s) = 6.4$ m, and $\Delta\psi = n \times 2\pi + 30^\circ$. Then $x(s) = 1$ mm.

Courant-Snyder Invariant

- In general,

$$x = A\sqrt{\beta} \sin \psi$$

$$x' = \frac{A}{\sqrt{\beta}} [\cos \psi - \alpha \sin \psi]$$

$$\beta x' = A\sqrt{\beta} [\cos \psi - \alpha \sin \psi]$$

$$= A\sqrt{\beta} \cos \psi - \alpha x$$

$$\boxed{\beta x' + \alpha x = A\sqrt{\beta} \cos \psi}$$

$$x^2 + (\beta x' + \alpha x)^2 = A^2 \beta$$

$$A^2 = \frac{x^2 + (\beta x' + \alpha x)^2}{\beta}$$

$$= \frac{x^2 + \alpha^2 x^2 + 2\alpha \beta x x' + \beta^2 x'^2}{\beta}$$

$$\boxed{A^2 = \gamma x^2 + 2\alpha x x' + \beta x'^2}$$

While C-S parameters evolve along the beam line, the combination above remains constant.

Properties of the Phase Space Ellipse

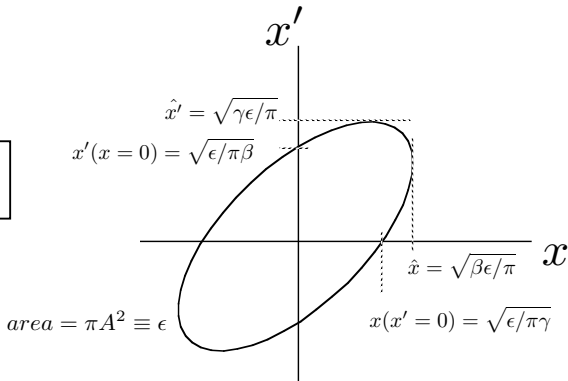
- The eqn. for the C-S invariant is that of an ellipse.
- If compute the area of the ellipse, find that ...

$$area = \pi A^2$$

i.e., while the ellipse changes shape along the beam line, its area remains constant

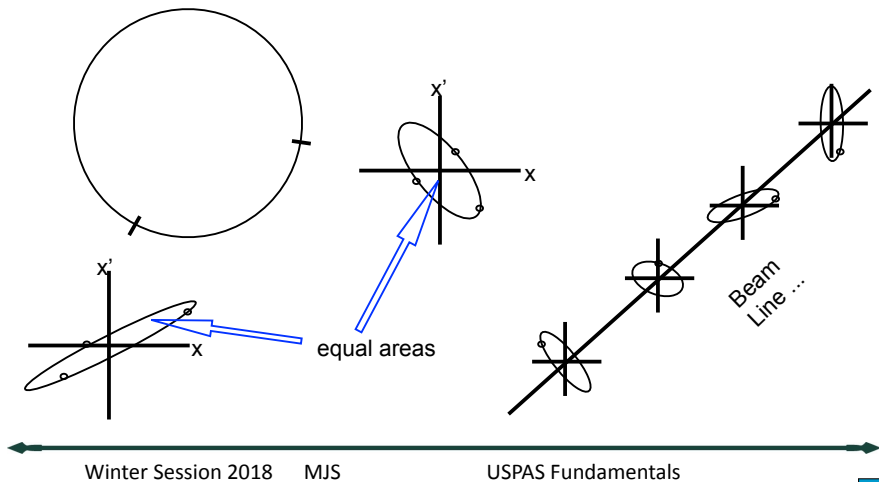
Emittance = area within a phase space trajectory

$$\gamma x^2 + 2\alpha x x' + \beta x'^2 = A^2$$



Motion in Phase Space

- Follow phase space trajectory...



Courant-Snyder Parameters

- So, we see that the solution to Hill's Equation leads to an amplitude function $\beta(s)$ and a related phase advance $\psi(s)$ such that $d\psi(s)/ds = 1/\beta(s)$. The general solution is
 - $x(s) = A \sqrt{\beta(s)} \sin[\psi(s) + \delta]$
- We see that if we define functions $\alpha(s) = -1/2 d\beta(s)/ds$ and $\gamma(s) = (1+\alpha(s)^2)/\beta(s)$, then the phase space trajectory of a single particle lies on an ellipse given by
 - ▶ $\gamma(s) x^2 + 2 \alpha(s) x x' + \beta(s) x'^2 = A^2$
- AND, the area of the ellipse is given by πA^2
- THUS, α, β, γ are the same parameters found to describe the phase space distributions discussed earlier



Choice of Initial Conditions

- Have seen how β , α can be propagated from one point to another. Still, have the choice of initial values...
- If periodic system, like a “ring,” then natural to choose the *periodic* solutions for β , α
- If beam line connects one ring to another ring, or a ring to a target, then we take the periodic solution of the upstream ring as the initial condition for the beam line
- In a system like a linac, wish to “match” to desired initial conditions at the input to the system (somewhere after the source, say) using an arrangement of focusing elements

Beam Mismatch

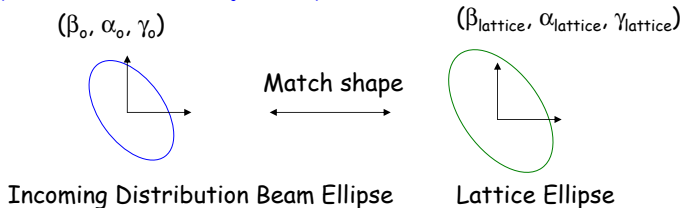


Beam Mismatch

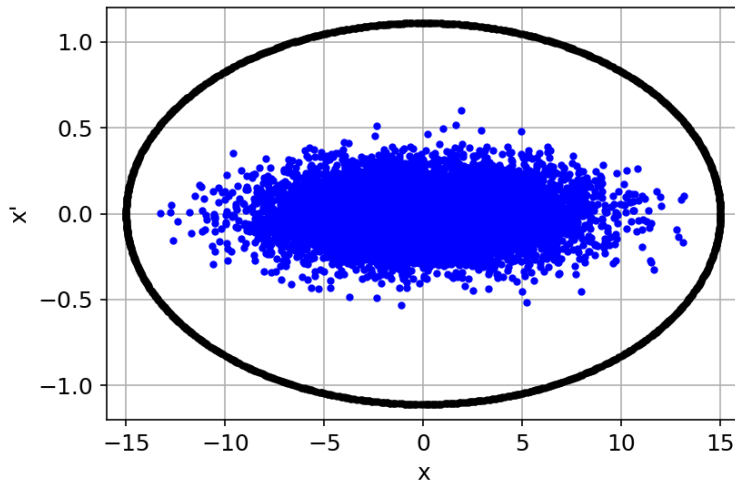
A closed, periodic system is a special case where we do not require an initial condition to calculate the beta function, $\beta(s)$. In this case, $\beta(s)$ is uniquely determined by the fact that we must require the initial and end conditions to match $\beta(s) = \beta(s+L_p)$. In the jargon of differential equations, transport lines are initial value problems, while rings have periodic boundary conditions.

For a periodic lattice ($K(s)=K(s+L_p)$), $\beta(s)$ is uniquely determined.

In any lattice, the phase space ellipses determined by the beam distribution should match those determined by the lattice Twiss parameters at the injection point:



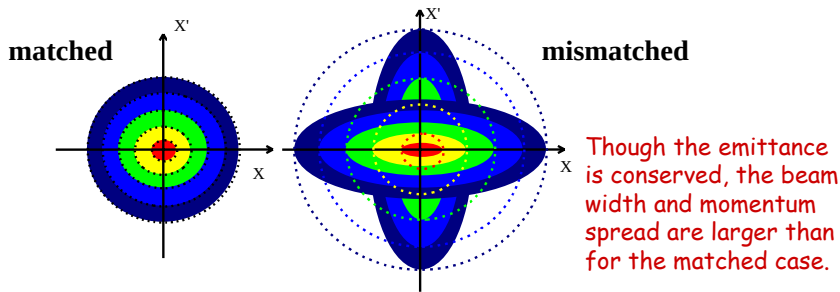
Beam mismatch demo





Beam Mismatch in a Periodic Lattice

If our initial beam does not match the Twiss parameters imposed by the lattice, we say that the beam is mismatched.



Mismatched beams do not return to the same Twiss ellipse turn-by-turn. Thus they are more likely to intercept the aperture and cause loss.

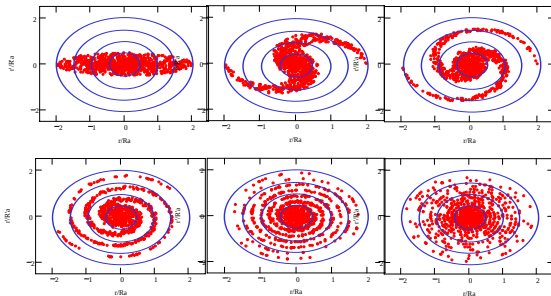


Effect of Mismatch in a Nonlinear Lattice

So far we have considered only the linear, homogeneous Hill's equation (we set all higher order terms to zero). In reality, some lattices have significant higher order terms.

$$u'' + K(s)u(s) = O[u^2] + \dots$$

When nonlinear terms are included, a mismatched beam will dilute in phase space, and emittance will grow..





Review of Transport Matrices

In the last two lectures, we have come up with a number of transport matrices, M . We now summarize these.

Matrix tracking (u, u') through a piecewise constant lattice:

$$M_n = \begin{pmatrix} \cos(\sqrt{K}l) & \frac{1}{\sqrt{K}}\sin(\sqrt{K}l) \\ -\sqrt{K}\sin(\sqrt{K}l) & \cos(\sqrt{K}l) \end{pmatrix} = \begin{pmatrix} C(s) & S(s) \\ C'(s) & S'(s) \end{pmatrix}$$

$$M_{total} = M_n M_{n-1} \dots M_1$$

Specific application to a FODO lattice, with focal lengths $ff=-fd=f$.

$$M_{FODO} = \begin{pmatrix} 1 - \frac{L^2}{2f^2} & 2L(1 + \frac{L}{2f}) \\ \frac{-L}{2f^2}(1 - \frac{L}{2f}) & 1 - \frac{L^2}{2f^2} \end{pmatrix}$$



Review of Transport Matrices

Matrix for tracking (u,u') through a lattice with K(s), in terms of β :

$$M = \begin{pmatrix} \sqrt{\frac{\beta}{\beta_o}} (\cos \varphi + \alpha_o \sin \varphi) & \sqrt{\beta \beta_o} \sin \varphi \\ \frac{\alpha - \alpha_o}{\sqrt{\beta \beta_o}} \cos \varphi - \frac{1 + \alpha_o \alpha}{\sqrt{\beta \beta_o}} \sin \varphi & \sqrt{\frac{\beta_o}{\beta}} (\cos \varphi - \alpha \sin \varphi) \end{pmatrix}$$

Specific application to a periodic lattice:.

$$M_{\text{Periodic}} = \begin{pmatrix} \cos \varphi + \alpha \sin \varphi & \beta \sin \varphi \\ -\gamma \sin \varphi & \cos \varphi - \alpha \sin \varphi \end{pmatrix}$$

And for a FODO with symmetry points at the end-points:

$$M_{\text{FODO}} = \begin{pmatrix} \cos \varphi & \beta \sin \varphi \\ -\frac{1}{\beta} \sin \varphi & \cos \varphi \end{pmatrix}$$



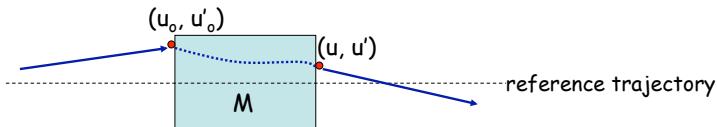
Matrix Representation of Motion

First take derivative: $u(s) = C(s)u_o + S(s)u_o'$

$$u'(s) = \frac{du}{ds} = C'(s)u_o + S'(s)u_o'$$

Then we can write the transport equation as a matrix:

$$\begin{pmatrix} u \\ u' \end{pmatrix} = \underbrace{\begin{pmatrix} C(s) & S(s) \\ C'(s) & S'(s) \end{pmatrix}}_M \begin{pmatrix} u_o \\ u_o' \end{pmatrix} = M \begin{pmatrix} u_o \\ u_o' \end{pmatrix}$$





Review of Transport Matrices

Finally, we can also transform the Twiss parameters themselves through the lattice:

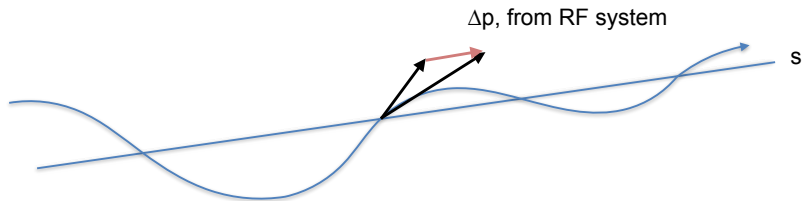
$$\begin{pmatrix} \beta \\ \alpha \\ \gamma \end{pmatrix} = \begin{pmatrix} C^2 & -2SC & S^2 \\ -CC' & (S'C + SC') & -SS' \\ C'^2 & -2S'C' & S'^2 \end{pmatrix} \begin{pmatrix} \beta_o \\ \alpha_o \\ \gamma_o \end{pmatrix}$$

where the C and S refer to the cosine-like and sine-like solutions found for the piece-wise constant lattice.

Normalized Emittance

Adiabatic Damping from Acceleration

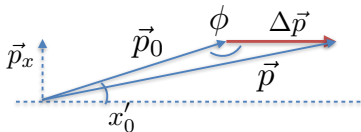
- Transverse oscillations imply transverse momentum. As accelerate, momentum is “delivered” in the longitudinal direction (along the s -direction). Thus, on average, the angular divergence of a particle will decrease, as will its oscillation amplitude, during acceleration.



- The coordinates $x-x'$ are not canonical conjugates, but $x-p_x$ are; thus, from classical mechanics, the area of a trajectory in $x-p_x$ phase space is invariant for adiabatic changes to the system.

Adiabatic Damping from Acceleration

details...



Note: assuming that ALL particles receive the same Δp from the cavity

$$x' = \frac{p_x}{p} = \frac{p_x}{\sqrt{p_0^2 + \Delta p^2 - 2\Delta p p_0 \cos \phi}} = \frac{p_x}{p_0} \left(1 - \frac{\Delta p}{p_0} + \dots \right) \approx x'_0 \left(1 - \frac{\Delta p}{p_0} \right)$$

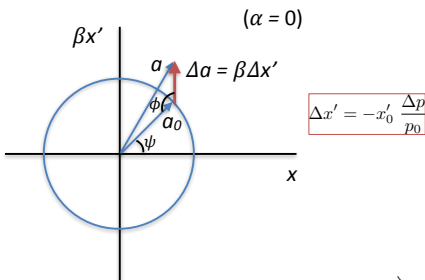
Note: particles at peak of their betatron oscillation will have little/no change in x' , while particles with large transverse angles will have their x' affected most

$$\Rightarrow \Delta x' = -x'_0 \frac{\Delta p}{p_0}$$

Adiabatic Damping from Acceleration

details...

Now assuming that the incremental change in momentum is small compared to the overall momentum, and that the changes occur gradually on time scales large compared to those of the betatron motion...



$$\begin{aligned}
 a^2 &= a_0^2 + \Delta a^2 - 2\Delta a a_0 \cos \phi \\
 &= a_0^2 + \Delta a^2 + 2\Delta a a_0 \sin \psi \\
 &= a_0^2 + \Delta a^2 + 2\Delta a \beta x'_0 \\
 &= a_0^2 + \left(-\beta x'_0 \frac{\Delta p}{p_0}\right)^2 + 2\left(-\beta x'_0 \frac{\Delta p}{p_0}\right)\beta x'_0 \\
 &= a_0^2 + (\beta x'_0)^2 \left(\frac{\Delta p}{p_0}\right)^2 - 2(\beta x'_0)^2 \frac{\Delta p}{p_0}
 \end{aligned}$$

$$\Rightarrow \langle a^2 \rangle = \langle a_0^2 \rangle - 2\langle (\beta x'_0)^2 \rangle \frac{\Delta p}{p_0}$$

Note: $2\langle (\beta x'_0)^2 \rangle = 2\langle x^2 \rangle = \langle a^2 \rangle$

So, $\Delta \langle a^2 \rangle = -\langle a^2 \rangle \frac{\Delta p}{p}$

$$\frac{\Delta \epsilon}{\epsilon} = -\frac{\Delta p}{p}$$

$$\epsilon \propto \frac{1}{p} \quad x_{rms} \propto \frac{1}{\sqrt{p}}$$

Normalized Beam Emittance

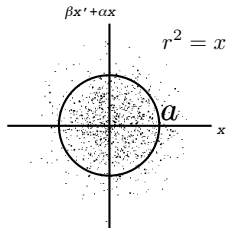
- Hence, as particles are accelerated, the area in x - x' phase space is not preserved, while area in x - p_x **is** preserved. Thus, we define a “normalized” beam emittance, as

$$\epsilon_N \equiv \epsilon \cdot (\beta\gamma)_{\text{Lorentz}}$$

- In principle, the *normalized* beam emittance should be preserved during acceleration, and hence along the chain of accelerators from source to target. Thus it is a measure of beam quality, and its preservation a measure of accelerator performance.
- In practice, it is not preserved -- non-adiabatic acceleration, especially at the low energy regime; non-linear field perturbations; residual gas scattering; charge stripping; field errors and setting errors; *etc.* -- all contribute at some level to increase the beam emittance. Best attempts are made to keep the emittance as small as possible.

Emittance of a Stationary Gaussian Distribution

- Imagine a transverse distribution of particles with a Gaussian profile in transverse coordinate x with zero mean and standard deviation σ .
 - The distribution can be described as follows:



$$r^2 = x^2 + (\beta x' + \alpha x)^2$$

$$\rho(r, \theta) r dr d\theta = \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2} r dr d\theta$$

$$\int_0^{2\pi} \int_0^a \rho r dr d\theta = f$$

Radius, a , containing fraction, f , of particles, corresponding to phase space area with emittance, ϵ :

$$a^2 = -2\sigma^2 \ln(1 - f) = \epsilon\beta/\pi$$

Emittance of a Stationary Gaussian Distribution

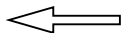
- So, the normalized emittance that contains a fraction f of a Gaussian beam is:

$$\epsilon_N = \frac{-2\pi \ln(1-f)\sigma^2(s)}{\beta(s)} (\beta\gamma)$$

← Lorentz!

- Common values of f :

f	$\epsilon_N / (\beta\gamma)$
95%	$6\pi\sigma^2 / \beta$
86.5%	$4\pi\sigma^2 / \beta$
39%	$\pi\sigma^2 / \beta$
15%	σ^2 / β



Perhaps most commonly used, typically called the “rms” emittance; but, always ask if not clear in context!

more typical for light sources, e- colliders

